Use: Autumn Science: Show Your Thinking

Today I can… ☐ Say how friction helps in one case. Say how it hinders in one case. ☐ Say what we gain from a mechanism. Say what it costs. ☐ For a fair test, say what to change. Say what to measure. Say what to keep the same. ☐ Explain an orbit. Say why we see light from the Moon.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Lesson strip

Arrival

I can explain familiar science using an observation, model or genuine piece of work.

In your book

In your book: Glue in your strip; answer four prompts.

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Arrival · Supported

1.

Choose: friction helps shoes grip or makes them float?

2.

Choose what the Moon orbits: Earth or the Sun.

3.

Keep the block the same or change it with the surface?

4.

Link to today: Choose whether the Moon reflects sunlight or makes its own light.

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Exit Ticket · Supported

1.

Pick one science idea. Point to the clue that backs it up.

In your book: Under the heading Exit ticket: answer the prompt; show one next step.

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Arrival · Standard

1.

Where does friction help, and where can it hinder?

2.

What is the pivot of a lever?

3.

Name two controls for a surface comparison.

4.

What keeps Earth in orbit, and what lights the Moon?

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Exit Ticket · Standard

1.

Pick one of the four science choices. Say why you made that choice. Use what was seen or the model to back it up.

In your book: Under the heading Exit ticket: answer the prompt; show one next step.

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Arrival · Stretch

1.

Give a case where friction helps. Name the two surfaces that touch.

2.

Use a lever. Explain how a gain in force links to the distance moved.

3.

You change the surface and the height you let go from. Why is it hard to know which change caused the result?

4.

How do the Moon's phases differ from a lunar eclipse?

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Exit Ticket · Stretch

1.

Look at the clues. Which is the least strong? Say what would make it stronger.

In your book: Under the heading Exit ticket: answer the prompt; show one next step.

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Starter and method strip

Starter

Two accounts say “it went farther”. Which one lets us explain why?

Method strip

1. Pick the cards you know. Use them for the four science choices.
2. Match each choice to what was seen. Stop and check the task with the adult. Check the help you have agreed.
3. Keep your first answer. Use just the help agreed. You do not need a new test.
4. Write the clues in your book. Say what the clues show in science. Make one change and check it.

Safe access

Genuine books, prepared evidence cards and familiar models.

Use the diagrams and cards you have. You do not need a new test. Use any models in the safe way agreed in class.

Put out just the kit for this stage.

If you cannot take part safely, watch the teacher show it. In your book, put what you saw. Say what it shows.

After the task, put what you saw in your book. Say what it shows about today's goal.

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You Do · Independent

Independent

Choose evidence for all four science decisions. Keep a first answer, use one private feedback point and improve one explanation.

Answer all four choices. Back each answer with a science reason.

I · S · C

Four science decisions

Where is friction helpful and where is it unhelpful? Name the touching surfaces.

Where is the pivot? If the effort force is smaller, what happens to effort distance?

Change the surface: what will you measure, and what stays the same?

Why does Earth orbit the Sun, and why does the Moon appear bright?

Use: Autumn Science: Show Your Thinking

Today I can… ☐ Say how friction helps in one case. Say how it hinders in one case. ☐ Say what we gain from a mechanism. Say what it costs. ☐ For a fair test, say what to change. Say what to measure. Say what to keep the same. ☐ Explain an orbit. Say why we see light from the Moon.

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You Do · Supported

Supported

Choose evidence for all four science decisions. Keep a first answer, use one private feedback point and improve one explanation.

Use prepared cards for friction, a lever, a fair comparison and Moonlight.

Point to the evidence, then finish one because statement.

Name the touching surfaces; gain and cost; control; light or orbit cause.

I · S · C

Four science decisions

Where is friction helpful and where is it unhelpful? Name the touching surfaces.

Where is the pivot? If the effort force is smaller, what happens to effort distance?

Change the surface: what will you measure, and what stays the same?

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You Do · Choice

Choice

Choose evidence for all four science decisions. Keep a first answer, use one private feedback point and improve one explanation.

Answer all four decisions using the evidence cards. Choose: friction helps shoes grip / makes shoes float; friction at a rubbing axle helps / wastes energy. Choose the pivot: turning point / end of the lever; less effort force needs more / less effort distance. For a surface comparison, choose measure distance / colour and keep the block same / different. Choose gravity / reflected light keeps Earth in orbit; the Moon reflects sunlight / makes light. Finish one because reason for each decision. Record C beside your book response.

I · S · C

Four science decisions

Where is friction helpful and where is it unhelpful? Name the touching surfaces.

Where is the pivot? If the effort force is smaller, what happens to effort distance?

Change the surface: what will you measure, and what stays the same?

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You Do · Aspire

Aspire

Choose evidence for all four science decisions. Keep a first answer, use one private feedback point and improve one explanation.

Explain each choice. Say how far a claim from a model or the data can go.

Say what the clues can show. Say one thing they cannot show.

I · S · C

Four science decisions

Where is friction helpful and where is it unhelpful? Name the touching surfaces.

Where is the pivot? If the effort force is smaller, what happens to effort distance?

Change the surface: what will you measure, and what stays the same?

Why does Earth orbit the Sun, and why does the Moon appear bright?

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Knowledge organiser

Learning goal

I can explain familiar science using an observation, model or genuine piece of work.

Blue shows word meanings. Green shows key ideas.

Word and meaning

friction — a force that acts against a slide, or a push to start a slide, where two surfaces touch

pivot — the point a lever turns around

control — something kept the same in a comparison

orbit — a curved path around another body

Key ideas

Use genuine work if it is available; a missing record stays missing.

Here are data to think about. The same block went 45 cm on smooth board. It went 19 cm on fabric. These data are here as an example. They are not results from your own test.

The evidence can support a claim about these surfaces and this block; one example is not every surface.

A lever may reduce the effort force by increasing the effort distance. It does not create energy.

Keep the claim fair

Illustrative data support reasoning practice only. They are never recorded as a completed practical.

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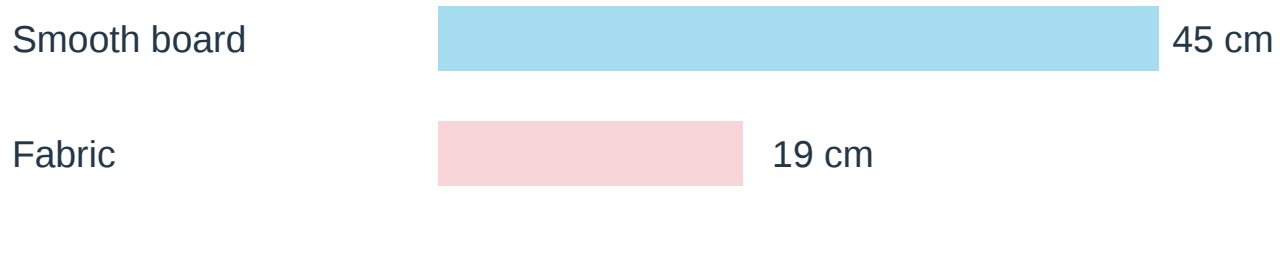
Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Knowledge organiser · continued

Comparison

Here are data to think about. The same block went 45 cm on smooth board. It went 19 cm on fabric. These data are here as an example. They are not results from your own test.

Blue shows smooth board. Pink shows fabric.



Illustrative data

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Word page

Forces and mechanisms

Blue shows forces and mechanisms. Green shows fair tests and orbits.

friction — a force that acts against a slide, or a push to start a slide, where two surfaces touch

pivot — the point a lever turns around

Fair comparison and orbits

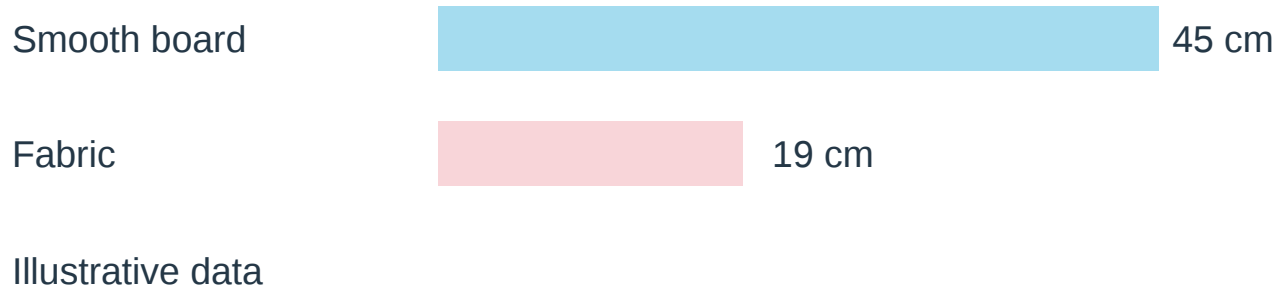
control — something kept the same in a comparison

orbit — a curved path around another body

Comparison

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Illustrative data

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We Do · response cards

Response cards

A B C

We Do

Which comparison is fair?

Change both surface and release height

Keep the block and release the same; change the surface

Keep only the colour the same

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We Do · cards and recording

We Do

Choose the destination for each card. Explain the evidence clue.

Read the card.

Choose its destination.

Check the reason and repair.

Keep one explanation in your book.

Brakes

Pads oppose sliding at the wheel.

Lever

The hand travels farther when less effort is needed.

Controls

Same block, release point and ramp; change the surface.

Destinations

Friction

Mechanisms

Fair comparison

Earth and Moon

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We Do · cards and recording · continued

Guided example

In your book: Record one placement and its science reason.

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We Do · cards and recording

We Do

Choose the destination for each card. Explain the evidence clue.

Read the card.

Choose its destination.

Check the reason and repair.

Keep one explanation in your book.

Moonlight

Sunlight reaches the Moon and reflects towards us.

Orbit

An inward pull changes a planet’s direction.

Engine

Unwanted rubbing produces heat.

Destinations

Friction

Mechanisms

Fair comparison

Earth and Moon

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We Do · cards and recording · continued

Guided example

In your book: Record one placement and its science reason.